

Path-integral measure for topological field theories

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We discuss how the dependence of the path-integral measure on the metric affects the properties of topological quantum field theories. We show that the choice of an invariant measure (under general coordinate transformations) preserves the topological character of these theories. We also discuss how topological invariants should be computed within this approach.

Topological quantum field theories (TQFTs), introduced by Witten [1–3] have received much attention [4–17]. One of the fundamental reasons to sustain the interest in these theories is their relevance to topology: since TQFTs are metric independent, they can only describe topological properties of the manifold on which they are defined so they are likely to provide a way to evaluate topological invariants via quantum field theory techniques. Topological models may also prove to be valuable tools for understanding fundamental physics; in particular three-dimensional (Chern–Simons) TQFT can be related to rational conformal field theories in two dimensions [3–14].

We address in this letter a problem already noted by Witten in his first work on TQFTs [1], namely, the possible dependence of the path integral measure on the metric. Since the central properties of these theories rely on the metric independence of the generating functional, the definition of a correct (invariant) measure is an important point which, to our knowledge, has not been discussed yet.

In a broad sense, a TQFT can be defined as a quantum field theory with a metric independent partition

function Z . All observables are metric independent and hence TQFTs are a class of generally covariant field theories. The way in which diffeomorphism invariance is realized in TQFTs is nevertheless quite peculiar: instead of promoting the metric into a fluctuating variable (like in gravity) the theory only invokes forms that refer to no geometrical connection.

In TQFTs, there is a basic “topological symmetry” which is the largest local symmetry possible for the fields that describe the system. This symmetry is responsible for gauging away any dependence on local properties.

As observed in ref. [4], TQFTs can be constructed by starting from a metric independent classical action S_{cl} and then BRST fixing this topological invariance. The resulting quantum action S_q takes the form

$$S_q[\Phi, g^{\mu\nu}] = S_{cl}[\Phi] + \{Q, K(\Phi, g^{\mu\nu})\}, \quad (1)$$

where Φ denotes the collective field content, $g^{\mu\nu}$ is the metric, Q is the (metric independent) BRST operator and K is some functional of the fields which eventually introduces a metric dependence on S_q .

One can distinguish between two types of TQFTs:

(1) The Schwartz type theories [18] for which S_{cl} is non-trivial (excluding, for example, total deriva-

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tive lagrangians). Chern–Simons theory in three dimensions belongs to this class [3].

(2) The Witten type theories [1,2], for which the complete quantum action can be written as a BRST commutator, i.e.

$$S_q = \{Q, V(\Phi, g^{\mu\nu})\}. \quad (2)$$

Yang–Mills TQFT in four dimensions [1] belongs to this class.

At this stage of the discussion, the common feature which both classes share is the fact that the corresponding energy–momentum tensor can be written as a BRST commutator. In particular, for Witten type theories one has from eq. (2)

$$T_{\mu\nu} = -\frac{2}{\sqrt{g}} \frac{\delta}{\delta g^{\mu\nu}} \{Q, V\} \quad (3)$$

or

$$T_{\mu\nu} = \left\{ Q, -\frac{2}{\sqrt{g}} \frac{\delta V}{\delta g^{\mu\nu}} \right\}. \quad (4)$$

Note that, in passing from eq. (3) to eq. (4) commutation between BRST and metric variations has been used. Now,

$$-\frac{2}{\sqrt{g}} \frac{\delta \log Z}{\delta g^{\mu\nu}} = \langle T_{\mu\nu} \rangle \quad (5)$$

and due to eq. (4), the RHS in eq. (5) is zero. To prove this last statement one uses BRST invariance of the path integral measure, this leading to the vanishing of the vacuum expectation value (VEV) of any BRST variation [1]. Then, metric independence of Z follows,

$$\frac{\delta Z}{\delta g^{\mu\nu}} = 0. \quad (6)$$

As has already been signaled, a possible dependence of the path integral measure $D\Phi$ on the metric has been ignored while obtaining eq. (5) from eqs. (2)–(4). It is the purpose of this letter to address to this point following Fujikawa's proposal concerning the specification of the correct integration variables in the presence of a metric [19–22]. Indeed, in the path integral measure

$$\prod_x D\Phi(x), \quad (7)$$

$\prod_x$ is non-trivial in curved space just as the volume

element d^4x is transformed non-trivially. Thus, appropriate weights for each field entering in Φ have to be introduced so that the correct integration variables $\tilde{\Phi}$ to define an invariant measure:

$$\prod_x D\tilde{\Phi}(x) = \prod_{x'} D\Phi(x'). \quad (8)$$

One way to determine $\tilde{\Phi}(x)$ is to choose BRST symmetry associated with general coordinate transformations as preferred symmetry. A more intuitive way to obtain $\tilde{\Phi}(x)$ follows from considering gaussian path integrals in curved space for fields with different tensorial character (see ref. [19] for details). The correctness of eq. (8) once Fujikawa's viewpoint is adopted has been confirmed by several methods [19–23].

We shall consider for definiteness Yang–Mills TQFT but our results apply to any other Witten type theory. Following the Labastida–Pernici [5] construction of TQFTs one starts from a classical gaussian action

$$S_{cl} = \frac{1}{2} \int d^4x \sqrt{g} \times \text{Tr}[(F_{\alpha\beta}^\dagger - G_{\alpha\beta}^\dagger)(F^{\dagger\alpha\beta} - G^{\dagger\alpha\beta})], \quad (9)$$

where

$$F_{\alpha\beta}^\dagger = \frac{1}{2}(F_{\alpha\beta} + \frac{1}{2}\epsilon_{\alpha\beta\gamma\delta}F^{\gamma\delta}), \quad (10)$$

$$F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha + i[A_\alpha, A_\beta].$$

Here, A_α is a gauge field taking values in the Lie algebra of the gauge group, $G_{\alpha\beta}^\dagger$ is an auxiliary anti-symmetric, self-dual field and $g = \det(g_{\mu\nu})$.

Action (9) is invariant under arbitrary variations of A_α provided an adequate variation of $G_{\alpha\beta}^\dagger$ is chosen. This is the large topological symmetry which has to be BRST fixed. One arrives a quantum action ([5–8]) given by

$$S_q = \{Q, V\} \quad (11)$$

with

$$V = \int d^4x \sqrt{g} \times \text{Tr}[\frac{1}{4}\chi_{\alpha\beta}(F^{\dagger\alpha\beta} - d^{\alpha\beta}) + \frac{1}{2}\lambda D_\alpha \psi^\alpha + b\partial^\alpha A_\alpha]. \quad (12)$$

The BRST transformations read

$$\{Q, A_\mu\} = i\psi_\mu - D_\mu c, \quad (13)$$

$$\begin{aligned}
\{Q, \chi_{\alpha\beta}\} &= d_{\alpha\beta} + \{c, \chi_{\alpha\beta}\}, \\
\{Q, \phi\} &= \{c, \phi\}, \\
\{Q, d_{\alpha\beta}\} &= [c, d_{\alpha\beta}] + i[\phi, \chi_{\alpha\beta}], \\
\{Q, \psi_\alpha\} &= -D_\alpha \psi + \{c, \psi_\alpha\}, \\
\{Q, \lambda\} &= 2i\eta + [c, \lambda], \\
\{Q, c\} &= (cc - i\phi), \\
\{Q, b\} &= d, \\
\{Q, d\} &= 0.
\end{aligned} \tag{13 cont'd}$$

Here, ψ_μ is the (Grassmann odd) ghost field associated with the topological symmetry, c is the (Grassmann odd) ghost field associated with the usual Yang-Mills gauge invariance and ϕ (Grassmann even) is the ghost of ghost corresponding to the second generation gauge invariance. Fields $d_{\alpha\beta}$, d and η are Lagrange multipliers (Grassmann even, even and odd respectively) and fields $\chi_{\alpha\beta}$, b and λ (Grassmann odd, odd and even respectively) are the usual anti-ghosts enforcing the gauge conditions. $d_{\alpha\beta}$ and $\chi_{\alpha\beta}$ are anti-symmetric and self-dual fields. All derivatives above should be taken as covariant derivatives in respect of general coordinate transformations.

The generating functional for Yang-Mills TQFT reads

$$Z = \int D\Phi \exp(-\{Q, \hat{V}[\Phi, g^{\mu\nu}]\}), \tag{14}$$

where, as announced, an invariant measure in terms of Fujikawa variables has been introduced.

Indeed, Fujikawa [19] has been able to completely specify an invariant local path integral measure in curved space. As we stated above, this can be achieved in two ways. One can impose the condition that the jacobian for BRST transformations associated with general coordinate transformations must be trivial. This requirement implies that the correct integration variables are not the original field Φ_i but appropriate tensorial densities $\hat{\Phi}_i$ constructed by multiplying the original fields Φ_i by adequate powers of g depending on their tensorial character:

$$\hat{\Phi}_i = g^{-w_i/2} \Phi_i, \tag{15}$$

where w_i is the weight associated with the field Φ_i . (There is an alternative way of obtaining the same result in the framework of the basic path integral for-

malism without recourse to the BRST symmetry, see refs. [19,20] for details.)

In Fujikawa's approach [19], the energy-momentum tensor is computed from Z by taking Φ and $g^{\mu\nu}$ as independent variables:

$$\langle \hat{T}_{\mu\nu} \rangle_\Phi = -\frac{2}{\sqrt{g}} \frac{\delta \log Z}{\delta g^{\mu\nu}}, \tag{16}$$

$$Z = \int D\Phi \exp(-\hat{S}[\Phi, g^{\mu\nu}]). \tag{17}$$

Since $D\Phi$ is a metric independent measure, all dependence on the metric is in $\hat{S}[\Phi, g^{\mu\nu}]$. One can then write

$$\begin{aligned}
\langle \hat{T}_{\mu\nu} \rangle_\Phi &= \frac{1}{Z} \int D\Phi \frac{2}{\sqrt{g}} \frac{\delta \hat{S}[\Phi, g^{\rho\sigma}]}{\delta g^{\mu\nu}} \\
&\times \exp(-\hat{S}[\Phi, g^{\rho\sigma}]).
\end{aligned} \tag{18}$$

Concerning the tensorial character of each field, it is determined by inspection of the transformation properties of the fields under general coordinate transformations and they, in turn, follow from the requirement that the action must be a general coordinate scalar.

Once the action (11) is explicitly written in terms of the fields defining the system, a straightforward analysis shows that the fields c , b , λ , d , η and ϕ are scalars. Fujikawa has shown that $w_{\text{scalar}} = -\frac{1}{2}$ and then, the correct variables are

$$\begin{aligned}
\{\hat{c} = g^{1/4}c, \hat{b} = g^{1/4}b, \hat{\lambda} = g^{1/4}\lambda, \hat{d} = g^{1/4}d, \\
\hat{\eta} = g^{1/4}\eta, \hat{\phi} = g^{1/4}\phi\}.
\end{aligned} \tag{19}$$

Concerning vectors A_μ and ψ_μ , their weights can be easily evaluated by using the vielbein e_μ^a to go over to flat indices. Indeed, writing $A_a = e_\mu^a A_\mu$ one can use the result for scalar fields:

$$D(A_a) = D(g^{1/4}e_\mu^a A_\mu) \tag{20}$$

or, returning to a coordinate basis,

$$D(\hat{A}_\mu) \equiv D(g^{1/8}A_\mu). \tag{21}$$

The same holds for ψ_μ and hence the correct variables are

$$\{\hat{A}_\mu = g^{1/8}A_\mu, \hat{\psi}_\mu = g^{1/8}\psi_\mu\}. \tag{22}$$

In the case of anti-symmetric self-dual fields $\chi_{\alpha\beta}$ and $d_{\alpha\beta}$ a similar analysis (taking into account that they have only three independent components) yields

$$\{\hat{\chi}_{\alpha\beta} = g^{-1/12} \chi_{\alpha\beta}, \hat{d}_{\alpha\beta} = g^{-1/12} d_{\alpha\beta}\}. \quad (23)$$

Once the correct integration variables $\hat{\Phi}$ have been identified, one has to express action S_q (11), (12) in terms of them. The corresponding generating functional then reads

$$Z = \int D\hat{\Phi}_i \exp(-\{Q, \hat{V}(\hat{\Phi}, g)\}), \quad (24)$$

where

$$\begin{aligned} \hat{V}(\hat{\Phi}, g^{\mu\nu}) = & \int d^4x \sqrt{g} \text{Tr} [\frac{1}{4} g^{1/12} g^{\alpha\rho} g^{\beta\sigma} \hat{\chi}_{\rho\sigma} \hat{F}_{\alpha\beta}^\dagger(\hat{A}_\epsilon) \\ & - \frac{1}{4} g^{1/6} g^{\alpha\rho} g^{\beta\sigma} \hat{\chi}_{\rho\sigma} \hat{d}_{\alpha\beta} \\ & + \frac{1}{2} g^{-1/4} \hat{\chi}^{\alpha\beta} \hat{D}_\beta(\hat{A}_\epsilon) (g^{-1/8} \hat{\psi}_\alpha) \\ & + g^{-1/4} \hat{b} g^{\alpha\beta} \partial_\beta (g^{-1/8} \hat{A}_\alpha)]. \end{aligned} \quad (25)$$

Using eq. (16) and eq. (18) the energy-momentum tensor can be computed. One can then write

$$\begin{aligned} \langle T_{\mu\nu} \rangle = & \frac{1}{Z} \int D\hat{\Phi} \frac{2}{\sqrt{g}} \frac{\delta}{\delta g^{\mu\nu}} \{Q, \hat{V}[\hat{\Phi}, g^{\rho\sigma}]\} \\ & \times \exp(-\{Q, \hat{V}[\hat{\Phi}, g^{\rho\sigma}]\}). \end{aligned} \quad (26)$$

It is important to stress that in terms of $\hat{\Phi}$ variables, commutation between BRST and metric variations does not hold. For instance, the BRST variation for $\hat{A}_\mu$ reads

$$\begin{aligned} \{Q, \hat{A}_\mu\} \\ = i\hat{\psi}_\mu - g^{1/8} [\partial_\mu (g^{-1/4} \hat{c}) + ig^{-3/8} [\hat{A}_\mu, \hat{c}]] \end{aligned} \quad (27)$$

which explicitly depends on the metric.

Then, it is not possible to interchange metric and BRST variations in eq. (26) so as to immediately conclude that the VEV of $T_{\mu\nu}$ is zero.

There is, however, a shift of fields with which this difficulty can be overcome. Indeed, Ouvry, Stora and van Baal found in ref. [7] a field redefinition which in terms of $\hat{\Phi}$ variables reads

$$\begin{aligned} \hat{\psi}'_\mu &= \hat{\psi}_\mu + ig^{-1/8} (\partial_\mu g^{-1/4} \hat{c} + ig^{-3/8} [\hat{A}_\mu, \hat{c}]), \\ \hat{\phi}' &= \hat{\phi} + \frac{1}{2} ig^{-1/4} \{\hat{c}, \hat{c}\}, \\ \hat{\eta}' &= \hat{\eta} - \frac{1}{2} ig^{-1/4} [\hat{c}, \hat{\lambda}], \\ \hat{d}'_{\alpha\beta} &= \hat{d}_{\alpha\beta} + g^{-1/4} \{\hat{c}, \hat{\chi}_{\alpha\beta}\} \end{aligned} \quad (28)$$

(all other variables remain unchanged).

With this shift of fields, BRST transformations read

$$\begin{aligned} \{Q, \hat{A}'_\mu\} &= i\hat{\psi}'_\mu, \quad \{Q, \hat{\psi}'_\mu\} = 0, \\ \{Q, \hat{\chi}'_{\alpha\beta}\} &= \hat{d}'_{\alpha\beta}, \quad \{Q, \hat{d}'_{\alpha\beta}\} = 0, \\ \{Q, \hat{c}'\} &= -i\hat{\phi}', \quad \{Q, \hat{\phi}'\} = 0, \\ \{Q, \hat{\lambda}'\} &= 2i\hat{\eta}', \quad \{Q, \hat{\eta}'\} = 0. \end{aligned} \quad (29)$$

From eq. (29) it follows that, in terms of the new variables, BRST and metric variations do commute.

We shall then consider (28) as a change of variables in the path integral defining Z . Since the associated jacobian can be easily seen to be one, we can write

$$Z = \int D\hat{\Phi}' \exp(-\{Q, \hat{V}'[\hat{\Phi}', g^{\rho\sigma}]\}) \quad (30)$$

with

$$\hat{V}'[\hat{\Phi}', g^{\rho\sigma}] = \int d^4x \sqrt{g} \text{Tr} \hat{v}'[\hat{\Phi}', g^{\rho\sigma}], \quad (31)$$

$\hat{v}'[\hat{\Phi}', g^{\rho\sigma}]$ being

$$\begin{aligned} \hat{v}'[\hat{\Phi}', g^{\rho\sigma}] = & \frac{1}{4} g^{1/12} g^{\alpha\mu} g^{\beta\nu} \hat{\chi}'_{\mu\nu} H_{\alpha\beta} \\ & + \frac{1}{2} g^{-1/4} g^{\alpha\beta} \hat{\chi}'_\beta(\hat{A}'_\epsilon) H_\alpha \\ & + g^{-1/4} g^{\alpha\beta} \hat{b}' \partial_\beta (g^{-1/8} \hat{A}'_\alpha), \end{aligned} \quad (32)$$

where

$$\begin{aligned} H_\alpha &= g^{-1/8} \hat{\psi}'_\alpha - i\partial_\alpha (g^{-1/4} \hat{c}') + g^{-3/8} [\hat{A}'_\alpha, \hat{c}'] \\ H_{\alpha\beta} &= \hat{F}_{\alpha\beta}^\dagger - g^{1/12} (\hat{d}'_{\alpha\beta} - g^{-1/4} \{\hat{c}', \hat{\chi}'_{\alpha\beta}\}). \end{aligned} \quad (33)$$

Now, instead of eq. (26) for $\langle T_{\mu\nu} \rangle$ we can write an expression where metric and BRST variations have been commuted:

$$\begin{aligned} \langle \hat{T}_{\mu\nu} \rangle_{\hat{\Phi}'} = & \frac{1}{Z} \int D\hat{\Phi}' \left\{ Q, \frac{2}{\sqrt{g}} \frac{\delta \hat{V}'}{\delta g^{\mu\nu}} \right\} \\ & \times \exp(-\{Q, \hat{V}'\}). \end{aligned} \quad (34)$$

From this, using the BRST invariance of the measure, one has

$$\langle \hat{T}_{\mu\nu} \rangle_{\hat{\Phi}'} = 0. \quad (35)$$

Then, taking into account the dependence of the path integral measure on the metric, the topological character of Witten type theories can be proven. There is, however, a difference between the naively computed $T_{\mu\nu}$ and the energy-momentum tensor obtained using the Fujikawa variables:

$$\hat{T}_{\mu\nu}[\hat{\Phi}', g^{\rho\sigma}] = T_{\mu\nu}[\hat{\Phi}', g^{\rho\sigma}] + A_{\mu\nu}[\hat{\Phi}', g^{\rho\sigma}], \quad (36)$$

with $A_{\mu\nu}$ still the BRST transformed of some functional.

The actual calculation of $\hat{T}_{\mu\nu}$ is straightforward although tedious. One finds

$$\begin{aligned} T_{\mu\nu}[\hat{\Phi}', g^{\rho\sigma}] = & \left\{ Q, \frac{1}{\sqrt{g}} \int d^4y \sqrt{g} \text{Tr} \left[\frac{1}{2} g_{\mu\nu} \hat{v}' [\hat{\Phi}', g^{\rho\sigma}] \right. \right. \\ & + g^{1/12} \hat{\chi}'_{\mu\rho} H^\rho_\nu + \frac{1}{2} g^{-1/4} \hat{\lambda}'_\mu \hat{D}'_\mu (\hat{A}_\epsilon) H_\nu \\ & + g^{-1/4} \hat{b}'_\mu (g^{-1/8} \hat{A}'_\nu) + \frac{1}{8} g^{1/12} g_{\mu\nu} \chi_{\rho\sigma} H^{\rho\sigma} \\ & \left. \left. + (\mu \leftrightarrow \nu) \right] \delta(x-y) \right\} \end{aligned} \quad (37)$$

and

$$\begin{aligned} A_{\mu\nu}[\hat{\Phi}', g^{\rho\sigma}] = & \left\{ Q, \frac{1}{\sqrt{g}} \int d^4y \sqrt{g} \text{Tr} \left[\frac{1}{2} g^{1/12} \hat{\chi}'_{\mu\rho} \right. \right. \\ & \times \left(\frac{1}{12} g_{\mu\nu} H^{\rho\sigma} \delta(x-y) + \frac{1}{2} \frac{\delta H^{\rho\sigma}(y)}{\delta g^{\mu\nu}(x)} \right) \\ & + g^{-1/4} \hat{\lambda}'_\mu \left(-\frac{1}{4} g_{\mu\nu} \delta(x-y) \hat{D}_\rho (\hat{A}_\epsilon) H^\rho \right. \\ & - \frac{1}{8} i g_{\mu\nu} g^{-1/8} \delta(x-y) [\hat{A}'_\rho, H^\rho] + \hat{D}_\rho (\hat{A}'_\epsilon) \frac{\delta H^\rho(y)}{\delta g^{\mu\nu}(x)} \\ & - \frac{1}{2} g^{-1/4} \hat{b}'_\mu \{ g_{\mu\nu} \delta(x-y) \partial^\rho (g^{-1/8} \hat{A}'_\rho) \\ & \left. \left. + \frac{1}{2} \partial^\rho [g_{\mu\nu} g^{-1/8} \delta(x-y) \hat{A}'_\rho] \right\} \right]. \end{aligned} \quad (38)$$

In TQFTs the expectation value of certain operators W corresponds to topological invariants provided the following conditions are fulfilled [1]: firstly, if the dimension $d(M)$ of the instanton moduli space M is not zero, the ghost number of W must equal $d(M)$; secondly, $\{Q, W\}$ must be zero with $W \neq \{Q, B\}$ for any B ; finally, $\delta W / \delta g^{\mu\nu} = 0$ (or, more generally, $\delta W / \delta g^{\mu\nu} = \{Q, \rho_{\mu\nu}\}$).

For example, for Yang-Mills TQFT, Witten [1] constructed several topological invariants starting from $W_0 = \frac{1}{2} \text{Tr} \phi^2$. From the arguments presented above it is evident that taking into account the dependence of the measure on the metric amounts to constructing operators from fields $\hat{\Phi}$ or, more conveniently, in terms of primed variables $\hat{\Phi}'$.

In particular, for the Yang-Mills case one could consider instead of W_0 referred to above,

$$W_0[\hat{\Phi}', g] = \frac{1}{2} \text{Tr} (\hat{\Phi}' - \frac{1}{2} i g^{-1/4} \{\hat{c}', \hat{c}'\})^2. \quad (39)$$

Although this operator is explicitly metric dependent, its variation in respect of the metric can be written as the BRST transform of some functional

$$\frac{\delta W_0}{\delta g^{\mu\nu}} = -\text{Tr} \frac{1}{12} g_{\mu\nu} g^{-1/4} \{\hat{Q}, \hat{c}' \hat{c}'\}, \quad (40)$$

and then, it leads to a topological invariant:

$$\begin{aligned} \langle W_0[\hat{\Phi}'] \rangle_{\hat{\Phi}'} = & \int D\hat{\Phi}' \text{Tr} \frac{1}{2} (\hat{\Phi}' - g^{1/4} [\hat{c}, \hat{c}])^2 \\ & \times \exp(-\{Q, \hat{V}'[\hat{\Phi}', g]\}). \end{aligned} \quad (41)$$

Other topological invariants can be constructed in the same way, following ref. [1].

In summary, the appropriate specification of the path integral variables in curved space, previously used for gravity, supergravity and string theory leading to a correct expression for gravitational and Weyl anomalies [19–22], can also be applied to the study of topological field theories. Choosing as an example Yang-Mills TQFT we have proven that the definition of an invariant measure (under general coordinate transformations) maintains the topological character of the theory originally shown just considering the quantum action S_q [1]. The explanation of our results lies in the fact that in Fujikawa's approach [19] the dependence of the measure on the metric is transferred to the quantum action S_q by an appropriate choice of independent variables. A fundamental property of Yang-Mills TQFT is $S_q = \{Q, V\}$ and this relation of course holds in terms of the correct variables $\hat{\Phi}$. Once one realizes this fact, one easily proves that $\delta Z / \delta g^{\mu\nu} = 0$ provided metric and BRST variations commute. One can find a change of variables leading to BRST transformations satisfying this property and then, the topological character of Yang-Mills theory follows immediately (of course, it should be also possible to show that $\delta Z / \delta g^{\mu\nu} = 0$ in terms of $\hat{\Phi}$ variables but this implies a more involved calculation). Since the features described above are shared by all Witten type TQFTs our results should hold for all of them. We have also explained how topological invariants must be computed in terms of the correct variables.

Finally, let us note that for Schwartz type TQFTs

(for example, the three-dimensional Chern–Simons model) S_q cannot be written as a BRST commutator then, only an explicit calculation of $\langle T_{\mu\nu} \rangle$ taking into account its regularization would show whether Z is still metric independent or not. We hope to report on this problem elsewhere.

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